

Muhammad Hasnain

PhD Candidate, Mechanical Engineering | J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, College Station, TX
mhasnain5924117@tamu.edu | [mhasnain5924117.github.io](https://github.com/mhasnain5924117) | [Google Scholar](#) | ORCID 0009-0004-6871-9713 | [GitHub](#)

RESEARCH INTERESTS

Scientific machine learning for transport phenomena; geometry-aware graph neural surrogates; physics-regularized learning; computational fluid dynamics and heat transfer; finite-volume modeling of reactive porous media; hydrogen storage; fire safety engineering; hybrid solver-ML workflows.

EDUCATION

PhD, Mechanical Engineering, Texas A&M University, 2024–present (expected 2027). Advisor: Prof. N. K. Anand.
Dissertation research: physics-regularized, geometry-generalizable graph neural surrogates for fluid flow and heat transfer.

MS, Mechanical Engineering, Georgia Southern University, 2022–2024. Advisor: Dr. Hayri Sezer. Thesis: Mathematical modeling of coupled heat and mass transfer in metal hydride hydrogen storage systems.

BS, Mechanical Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI), Pakistan, 2017–2021. Exchange semester, Western Carolina University (Spring 2020), U.S. Department of State Global UGRAD scholar.

RESEARCH EXPERIENCE

Graduate Research Assistant, Computational Heat Transfer Laboratory, Texas A&M University, Aug 2024–present.
Graph neural network surrogates on unstructured CFD meshes (PyTorch Geometric); physics-regularized loss design for geometry generalization; contribution to the ShockCast adaptive time-stepping framework (ICLR 2026); method-of-manufactured-solutions data generation for neural-operator benchmarking; OpenFOAM coupling.

Graduate Research Assistant, Fire and Energy (FireGY) Laboratory, Georgia Southern University, Aug 2022–Jul 2024.
In-house finite-volume solvers (Python, MATLAB) for coupled heat, mass, and momentum transfer with reaction kinetics; metal hydride hydrogen storage reactors; three-dimensional gypsum board dehydration model validated against cone-calorimeter data (NIJ-funded project); lithium-ion thermal runaway and fire-induced tree injury models; PINNs in Julia SciML.

JOURNAL ARTICLES

M. Hasnain, H. Sezer, S. P. Kozhumal. Predicting thermal response of gypsum board under various heat flux configurations: A three-dimensional mathematical model. *International Journal of Thermal Sciences* 221, 110457, 2026. doi:10.1016/j.ijthermalsci.2025.110457

M. Hasnain, H. Sezer, J. H. Mason. Mathematical modeling of heat and mass transfer in metal hydride hydrogen storage systems: A comprehensive review. *Renewable and Sustainable Energy Reviews* 226, 116294, 2026. doi:10.1016/j.rser.2025.116294

M. Hasnain, N. K. Anand, A. Srinivasa, P. Kumar. A novel residual multilayer perceptron-based graph convolutional network for modeling heat transfer and fluid flow. *Physics of Fluids* 37(10), 107138, 2025. doi:10.1063/5.0292956

M. Hasnain, H. Sezer, J. H. Mason. Modeling heat and mass transfer in metal hydride hydrogen storage systems: Impact of operating parameters and reactor geometry. *International Journal of Hydrogen Energy* 71, 1045–1055, 2024. doi:10.1016/j.ijhydene.2024.05.311

REFEREED CONFERENCE PAPERS

J. Helwig, S. S. Adavi, X. Zhang, Y. Lin, F. S. Chim, L. T. Vizzini, H. Yu, **M. Hasnain**, S. K. Biswas, J. J. Holloway, N. Singh, N. K. Anand, S. Guhathakurta, S. Ji. A two-phase deep learning framework for adaptive time-stepping in high-speed flow modeling. *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2506.07969

M. Hasnain, S. Khan, M. A. Ezazi, H. Sezer. Modeling heat and mass transfer in metal hydride-based hydrogen storage systems using the finite volume method. *Proc. ASME IMECE 2023*, V010T11A047. doi:10.1115/IMECE2023-112874

S. Khan, S. Anwar, J. Casa, **M. Hasnain**, H. Ahmed, H. Sezer. Modeling thermal runaway in prismatic lithium-ion batteries. *Proc. ASME IMECE 2023*, V010T11A048. doi:10.1115/IMECE2023-113787

PREPRINTS AND MANUSCRIPTS UNDER REVIEW

- M. Hasnain**, N. K. Anand, et al. A physics-regularized graph neural surrogate for geometry-generalizable fluid flow prediction. Under review, *Journal of Computational Physics*, 2026.
- M. U. Khan, **M. Hasnain**, H. Sezer, S. P. Kozhumal, M. K. Selvaraj, A. Simeoni. Predictive heat transfer model for fire-induced injuries in tree stem. Research Square preprint, 2024; under review, *International Communications in Heat and Mass Transfer*. doi:10.21203/rs.3.rs-5671657/v1

CONFERENCE PRESENTATIONS AND ABSTRACTS

- M. Hasnain**, N. K. Anand, et al. Physics-regularized graph neural surrogates for geometry-generalizable fluid flow prediction. WCCM-ECCOMAS 2026, Munich, Germany, Jul 2026.
- M. Hasnain**, N. K. Anand, et al. Application of graph neural networks for heat transfer and fluid flow problems. ASME IMECE 2025, Memphis, TN, Nov 2025.
- M. Hasnain**, R. Paye, T. Borth, G. Gorbett, S. P. Kozhumal, H. Sezer. Three-dimensional mathematical modeling of heat and mass transfer during calcination of gypsum board exposed to fire. ESSCI Spring Technical Meeting, Athens, GA, Mar 2024.
- S. Khan, **M. Hasnain**, et al. Mathematical modeling of heat-induced decomposition kinetics leading to thermal runaway in lithium-ion batteries. ESSCI Spring Technical Meeting, Mar 2024.
- M. U. Khan, **M. Hasnain**, S. Khan, et al. Mathematical model for predicting injuries and temperature distributions in stem exposed to fire. ESSCI Spring Technical Meeting, Mar 2024.
- A. Idris, S. Khan, **M. Hasnain**, et al. Numerical modeling of spontaneous ignition in large coal stockpiles using finite volume method. ESSCI Spring Technical Meeting, Mar 2024.
- M. Hasnain**, R. Paye, J. Casa, T. Borth, G. E. Gorbett, S. P. Kozhumal, H. Sezer. 3D mathematical model for heat and mass transfer mechanisms in gypsum board exposed to fire. 13th U.S. National Combustion Meeting, College Station, TX, Mar 2023.
- S. Khan, **M. Hasnain**, A. Liaqat, V. Akkerman, H. Ahmed, S. P. Kozhumal, H. Sezer. The self-heating ignition of lithium-ion batteries: Comparative study of COMSOL and GPYRO models. 13th U.S. National Combustion Meeting, Mar 2023.

POSTERS

- M. Hasnain**, H. Sezer. Mathematical modeling of coupled heat and mass transfer in metal hydrides for safer and more efficient hydrogen storage systems. Texas A&M Conference on Energy, Sep 2024.
- M. Hasnain**, M. U. Khan. Three-dimensional multiphysics modeling of heat and mass transfer during calcination of gypsum board for forensic fire investigations. Georgia Southern University Student Research Symposium, Apr 2024. Best Graduate Student (Top Eagle) Award.
- M. Hasnain**, H. Sezer. Modeling and optimization of metal hydride based hydrogen storage systems. Georgia Southern University Student Research Symposium, Apr 2023.
- M. Hasnain**, H. Sezer. Diffusion-based mathematical modeling of hyperbaric oxygen therapy for chronic wounds. Georgia Southern University Student Research Symposium, Apr 2023.
- S. Khan, **M. Hasnain**, H. Sezer. Mathematical modeling of thermal metamaterials. Georgia Southern University Student Research Symposium, Apr 2023.

INVITED TALKS

- Spotlight talk, RAISE AI Bridge Summit, Texas A&M University, Jan 2026.
- Invited session on research design and methodology, Undergraduate Research Group (UROG), GIKI, Apr 2026.

TEACHING AND MENTORING

- Graduate Teaching Assistant**, MEEN 344 Fluid Mechanics, Texas A&M University, Fall 2024. Delivered several full lectures, weekly office hours, and grading for 90+ undergraduates.
- Graduate Teaching Assistant / Laboratory Instructor**, Georgia Southern University, 2023–2024: MENG 4210 Energy Science Laboratory; ENGR 2139 Numerical Methods; ENGR 2112 Solid Modeling and Analysis; MENG 7136 Mechatronics I Laboratory.
- Mentoring**: trained undergraduate and graduate researchers in the FireGY Laboratory in finite-volume modeling (FVTool workshops on diffusion and diffusion-advection problems), several of whom co-authored posters and abstracts; peer

mentor, Mentor Collective, Georgia Southern; Math Expert, Photomath.

Pedagogical training: Academy for Future Faculty (in progress) and Teaching Assistant Institute (completed), Texas A&M Center for Teaching Excellence; CIRTl Network.

HONORS AND AWARDS

Graduate and Professional School Travel Awards (\$1,750), Texas A&M University, for WCCM-ECCOMAS 2026 (2026).

Graduate Student Travel Award (\$750), J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University (2025).

Best Graduate Student (Top Eagle) Award, Georgia Southern University Student Research Symposium, sponsored by the Georgia Center of Innovation (2024).

Georgia Power Innovation Award, second place, Georgia Southern University Student Research Symposium (2024).

Southern Automation, Logistics and Technology (SALT) Computing Award and Gulfstream Aerospace Modeling Award, Georgia Southern University Student Research Symposium (2023).

Graduate Student Organization Research Award (\$750) and Travel Grant for IMECE 2023, Georgia Southern University (2023).

Global UGRAD Scholar, U.S. Department of State; LEAP Institute, George Mason University (2020).

Khyber Pakhtunkhwa Chief Minister Merit Scholarship, GIKI (2017–2021).

RESEARCH PROPOSALS

Awarded: National Institute of Justice, thermo-chemistry model and property database development for fire-exposed wall-lining materials (\$290,289). PI: Hayri Sezer. Role: concept development, mathematical modeling, proposal writing.

Under review: DOE NEUP CINR, graph-neural-network-based optimization of nuclear reactor thermal-hydraulics. PI: N. K. Anand. Role: concept and methodology formulation, proposal development.

Under review: DOE NEUP CINR, efficient large-eddy simulation with learned closures. PI: N. K. Anand. Role: literature synthesis, conceptual framework, writing.

Submitted: NASA, multiphysics and multiscale modeling of in-space additive manufacturing. PI: Hayri Sezer. Role: draft preparation and model conceptualization.

SERVICE AND OUTREACH

Volunteer, Texas Science and Engineering Fair; volunteer, Spark! PK–12 Engineering Outreach, Texas A&M College of Engineering; peer mentor, Mentor Collective, Georgia Southern University; Fellow, Academy for Future Faculty, Texas A&M.

TECHNICAL SKILLS

Python, MATLAB, Julia, C++; PyTorch, PyTorch Geometric, TensorFlow, Julia SciML; OpenFOAM, in-house finite-volume solvers, COMSOL, ANSYS; SolidWorks, Fusion 360; validation with TGA/DSC and cone-calorimeter data.

REFERENCES

Prof. N. K. Anand (PhD advisor), J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University.

Dr. Hayri Sezer (MS advisor), Department of Mechanical Engineering, Georgia Southern University.

Contact details for references are provided with application materials.

Last updated September 2026.